

Statistics (7 th std term 3)

Choose the Appropriate Answer

1. The mean of first ten natural numbers is _____.
(a) 6.5 (b) 5.5 (c) 5 (d) 6
Ans: (b) 5.5
2. If the average selling price of 15 books is 235, then the total selling price is ----.
(a) 3525 (b) 3025 (c) 4025 (d) None
Ans: (a) 3525
3. The average of the marks 2, 9, 5, 4, 4, 8, 10 is _____.
(a) 5 (b) 7 (c) 8 (d) 6
Ans: (d) 6
4. The average of integers between -10 to 10 is _____.
(a) 10 (b) -10 (c) Zero (d) None
Ans: (c) Zero
5. _____ is a representative value of the entire data.
(a) Mean (b) Range (c) Minimum value (d) Maximum value
Ans: (a) Mean
6. The mean of first fifteen even numbers is _____.
(a) 4 (b) 16 (c) 5 (d) 10
Ans: (b) 16

$$15 \times (15 + 1) = 15 \times 16 = 240.$$
$$\text{The average is } 240 \div 15 = 16.$$

7. The average of two numbers are 20. One number is 24, another number is _____.
(a) 16 (b) 26 (c) 20 (d) 40
Ans: (a) 16
8. The mean of the data 12, x, 28 is 18. Find the value of x.
(a) 18 (b) 16 (c) 14 (d) 22
Ans: (c) 14

$$\text{Mean} = \frac{12 + x + 28}{3} = 18.$$
$$40 + x = 54, \text{ so } x = 14.$$

9. Find the mode of the following data. 2, 4, 5, 2, 6, 7, 2, 7, 5, 4, 8, 6, 10, 3, 2, 4, 2
(a) 5 (b) 2 (c) 6 (d) 7
Ans: (b) 2

10. The colors used by the six students for drawing is blue, orange, yellow, white, green and blue then the mode is _____.

- (a) Blue (b) Green (c) White (d) Yellow

Ans: (a) Blue

11. Find the mode of data 3, 6, 9, 12, 15

- (a) 1 (b) 2 (c) 3 (d) No mode

Ans: (d) No mode

12. Find the modes of the data 2, 1, 1, 3, 4, 5, 2.

- (a) 1 and 5 (b) 2 and 3 (c) 2 and 1 (d) 1 and 4

Ans: (c) 2 and 1

13. Find the mode of 2, 5, 5, 1, 3, 2, 2, 1, 3, 5, and 3.

- (a) 1, 2 and 3 (b) 1, 2 and 5 (c) 2, 3 and 5 (d) None

Ans: (c) 2, 3 and 5

14. Find the median of 3, 8, 7, 8, 4, 5, and 6.

- (a) 4 (b) 8 (c) 5 (d) 6

Ans: (d) 6

The numbers are arranged in ascending order: 3, 4, 5, 6, 7, 8, 8.

$$n = 7 \Rightarrow \frac{n+1}{2} \Rightarrow \frac{7+1}{2} = \frac{8}{2} = 4.$$

The 4th value in the ordered list is 6.

15. Find the median of 11, 14, 10, 9, 14, 11, 12, 6, 7, and 7.

- (a) 11.5 (b) 10.5 (c) 11 (d) 14

Ans: (b) 10.5

The numbers are arranged in ascending order: 6, 7, 7, 9, 10, 11, 11, 12, 14, 14.

$$\text{The median is calculated as } \frac{10+11}{2} = \frac{21}{2} = 10.5.$$

16. If the median of a, 2a, 4a, 6a, 9a is 8, then find the value of a is

- (a) 8 (b) 6 (c) 2 (d) 10

Ans: (c) 2

$$\text{The position is } \frac{5+1}{2} = \frac{6}{2} = 3 \Rightarrow \text{The 3}^{\text{rd}} \text{ value is } 4a \Rightarrow 4a = 8 \Rightarrow a = 2$$

17. The median of the data 24, 29, 34, 38, 35 and 30, is _____.

- (a) 29 (b) 30 (c) 34 (d) 32

Ans: (d) 32

18. The median first 6 odd natural numbers is _____.

- (a) 6 (b) 7 (c) 8 (d) 14

Ans: (a) 6

19. The median of first ten even natural numbers is _____.
(a) 12 (b) 11 (c) 10 (d) 9

Ans: (b) 11

The first ten even natural numbers are 2,4,6,8,10,12,14,16,18,20
The positions are $\frac{10}{2} = 5$ and $\frac{10}{2} + 1 = 6 \Rightarrow \frac{10 + 12}{2} = \frac{22}{2} = 11$.

20. A batsman scored the following number of runs in six innings: 35, 30, 45, 65, 39, and 20. The mean runs scored by him in an inning is
(a) 39 (b) 38 (c) 37 (d) 40

Ans: (a) 39

21. The mean of the numbers 10, 20, 30, and 40 is
(a) 20 (b) 25 (c) 30 (d) 50

Ans: (b) 25

22. The range of the weights (in kg) of students of a class given below is:
49, 60, 47, 50, 47, 59, 58, 45, 53
(a) 10 (b) 15 (c) 20 (d) 2

Ans: (b) 15

23. The mode of the distribution 3,5, 7, 4, 2, 1, 4, 3, 4 is
(a) 7 (b) 4 (c) 3 (d) 1

Ans: (b) 4

24. Median of distribution 2, 3, 4, 7, 5, 1 is:
(a) 4 (b) 7 (c) 11 (d) 3.

Ans: (a) 4

25. Mean of the first six prime numbers is
(a) 7.83 (b) 5.83 (c) 6.83 (d) 4.83

Ans: (c) 6.83

26. The data which is collected directly from the source is called
(a) Primary data (b) Ungrouped data (c) Grouped data (d) Secondary data

Ans: (a) Primary data

27. The mean of the first ten natural numbers is
(a) 6.5 (b) 5.5 (c) 7.5 (d) 8.5

Ans: (b) 5.5

28. What is called one or more outcomes of an experiment?
(a) Space (b) Experiment (c) Sample (d) Event

Ans: (d) Event

29. The probability of an experiment cannot be greater than :
(a) 0 (b) 0.5 (c) 1 (d) 2

Ans: (c) 1

30. The mean of 6, y, 7, x, and 14 is 8. Which of the following is true?
(a) $x + y = 13$ (b) $x - y = 13$ (c) $2x + 3y = 13$ (d) $x^2 + y = 1$

Ans: (a) $x + y = 13$

$$\text{Mean} = \frac{\text{number of observations}}{\text{total observation}} \Rightarrow 8 = \frac{6+y+7+x+14}{5} \Rightarrow 40 = 27+y+x \Rightarrow y+x = 13$$

31. The mean of the first five whole number is _____.
(a) 2 (b) 5 (c) 3 (d) 4

Ans: (a) 2

32. The mean of the first five natural number is _____.
(a) 2 (b) 5 (c) 3 (d) 4

Ans: (c) 3

33. The mean of the first seven natural number is _____.
(a) 2 (b) 5 (c) 3 (d) 4

Ans: (d) 4

34. The median of the first ten natural number is _____.
(a) 2.5 (b) 5.5 (c) 3.5 (d) 4.5

Ans: (b) 5.5

35. The median of the first ten prime number is _____.
(a) 2.5 (b) 5.5 (c) 3.5 (d) None of these

Ans: (d) None of these

36. A cricketer scores the following runs in eight innings: 58, 76, 40, 35, 46, 45, 0, and 100. What will be their mean score?
(a) 400 (b) 50 (c) 200 (d) 100

Ans: (b) 50

37. What will be the range of following data? 32, 41, 28, 54, 35, 26, 33, 23, 38, 40
(a) 25 (b) 23 (c) 31 (d) 54

Ans: (c) 31

38. Which observation in the following data has maximum frequency?
1, 1, 2, 4, 3, 2, 1, 2, 2, 4

(a) 4 (b) 3 (c) 1 (d) 2

Ans: (d) 2

39. The mode of the data 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 8 is _____.
(a) 2 (b) 5 (c) 8 (d) 2 & 5 both

Ans: (d) 2 & 5 both

40. A data can have _____ mode.
(a) Only one (b) Only two (c) Only 3 (d) More than one

Ans: (d) More than one

41. The tally mark **||||** shows frequency _____.
(a) 4 (b) 5 (c) 0 (d) 3

Ans: (b) 5

42. The tally mark **|||| |||| ||||** shows frequency _____.
(a) 15 (b) 13 (c) 12 (d) None of these

Ans: (a) 15

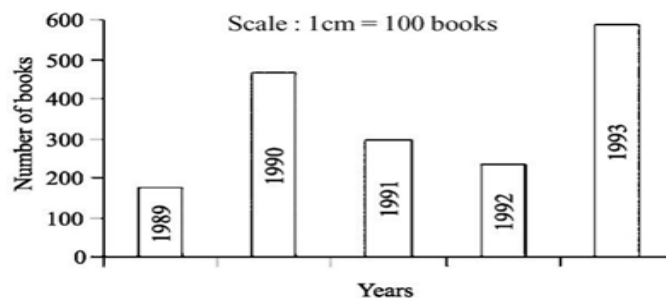
43. Median of the data 13, 16, 12, 14, 19, 12, 14, 13, 14 is _____.
(a) 14 (b) 19 (c) 12 (d) 13

Ans: (a) 14

44. Mode and median of the data 13, 16, 12, 14, 19, 12, 14, 13, 14 are:
(a) 13 & 14 (b) 14 & 13 (c) 14 & 14 (d) 19 & 13

Ans: (c) 14 & 14

Read the following bar graph which shows the number of books sold by a bookstore during five consecutive years and answer the question given below (45 - 50)



45. How many books were sold in 1989?
(a) 100 (b) 200 (c) 300 (d) 600

Ans: (b) 200

46. In which year were 300 books sold?
(a) 1990 (b) 1991 (c) 1993 (d) 1989

Ans: (b) 1991

47. In which year were fewer than 200 books sold?
(a) 1989 (b) 1991 (c) 1993 (d) 1992

Ans: (a) 1989

48. What will be the difference of number of books sold in 1993 and 1990?
(a) 600 (b) 200 (c) 400 (d) 100

Ans: (d) 100

49. How many books were sold from 1989 to 1991?
(a) 600 (b) 1000 (c) 400 (d) 800

Ans: (b) 1000

50. How many books were sold from 1991 to 1993?
(a) 1300 (b) 1000 (c) 900 (d) 800

Ans: (a) 1300

51. Median of the data 9,8,1,2,3,6,7,5,4 is ----

- (a) 5 (b) 9 (c) 6 (d) 4

Ans: (a) 5

52. Ashish studies for 4 hours, 5 hours and 3 hours respectively on three consecutive days. How many hours does he study daily on an average?

- (a) 5 (b) 9 (c) 6 (d) 4

Ans: (d) 4

53. A batsman scored the following number of runs in six innings: 36, 35, 50, 46, 60, and 55. Find the mean runs scored by him in an inning.

- (a) 50 (b) 49 (c) 46 (d) 47

Ans: (d) 47

54. The mean of the first five odd natural number is _____.

- (a) 2 (b) 5 (c) 3 (d) 4

Ans: (b) 5

55. The mean of the first ten odd natural number is _____.

- (a) 12 (b) 15 (c) 10 (d) 11

Ans: (c) 10

56. The mean of the first ten even natural number is _____.

- (a) 12 (b) 15 (c) 10 (d) 11

Ans: (d) 11

57. The median of the first ten even natural number is _____.

- (a) 12 (b) 15 (c) 10 (d) 11

Ans: (d) 11

58. The mean of the first ten prime number is _____.

- (a) 12.5 (b) 12.9 (c) 12.8 (d) 14.5

Ans: (b) 12.9

59. Find the mode of the given set of numbers: 1, 1, 2, 4, 3, 2, 1, 2, 2, 4

- (a) 2 (b) 1 (c) 3 (d) 4

Ans: (a) 2

60. The mode of the given set of numbers 14, 16, 12, 14, 14, 16, 14, 10, 14, 18, 14 is

- (a) 12 (b) 14 (c) 16 (d) 11

Ans: (b) 14

61. Find the mode of the following data: 12, 14, 12, 16, 15, 13, 14, 18, 19, 12, 14, 15, 16, 15, 16, 16, 15, 17, 13, 16, 16, 15, 15, 13, 15, 17, 15, 14, 15, 13, 15, 14

- (a) 12 (b) 14 (c) 16 (d) None of these

Ans: (d) None of these

The mode of the given data is 15

62. The range of the given data is 20, 19, 13, 12, 6, and 18.

- (a) 20 (b) 19 (c) 13 (d) 14

Ans: (d) 14

63. The range of data is also called

- (a) Spread (b) Mean (c) Value (d) Frequency

Ans: (a) Spread

64. Sum of the value of the data divided by the total no of observations is known as

- (a) Range (b) Mean (c) Median (d) Mode

Ans: (b) Mean

65. The middle value of data when it is arranged in ascending or descending order.

- (a) Range (b) Mean (c) Median (d) Mode

Ans: (c) Median

66. Median of the given data is 12, 8, 9, 10 and 6.

- (a) 6 (b) 9 (c) 10 (d) 12

Ans: (b) 9

67. The mode of the given data is 3, 4, 5, 6, 7, 5...

- (a) 3 (b) 7 (c) 5 (d) 4

Ans: (c) 5

68. Mode of the given data 5, 15, 25, 35, 45.....

- (a) 25 (b) 15 (c) 45 (d) None of these

Ans: (d) None of these

69. Weight of students in a class is 30 kg, 42 kg, 50 kg, 36 kg & 47 kg. The average weight is

- (a) 41 kg (b) 42 kg (c) 37.5 kg (d) 45.7 kg

Ans: (a) 41 kg

70. Median of the give data 6, 2, 3, 4, 5, 7 is

- (a) 6 (b) 2 (c) 10 (d) 4.5

Ans: (d) 4.5

71. The mean of first 5 multiples of 2 is

- (a) 5 (b) 6 (c) 10 (d) None of these

Ans: (c) 10

72. The relation between Mean (A), Median (B) and Mode (C) is

- (a) $C = 3B - 2A$ (b) $C = 2B - 3A$ (c) $C = 3A - 2B$ (d) $C = 2A - 3B$

Ans: (c) $C = 3A - 2B$

73. Mean of 8, 10, 12, x, 4 is 10. Then the value of x is

- (a) 14 (b) 16 (c) 12 (d) 6

Ans: (b) 16

74. Mean, median, and mode are measures of _____.

(a) Frequency (b) Central tendency (c) Variation (d) Probability
Ans: (b) Central tendency

75. The mean weight of 100 students in a class is 46 kg. The mean weight of boys is 50 and of girls is 40 kg. Therefore, the number of boys is:

- (a) 50 (b) 60 (c) 70 (d) 65

Ans: (b) 60

76. The number of times an observation occurs in a data is called its

- (a) Range (b) Raw data (c) Interval (d) Frequency

Ans: (d) Frequency

77. The mean of five numbers is 27. If one of the numbers is excluded, the mean gets reduced by 2. What is the excluded number?

- (a) 35 (b) 27 (c) 25 (d) 40

Ans: (a) 35

78. The arithmetic mean of five given numbers is 85. What is their sum?

- (a) 425 (b) A number between 85 and 425.
(c) 85 (d) A number greater than 500.

Ans: (a) 425

79. The mean of 6, y, 7, x and 14 is 8. Which of the following is true?

- (a) $x + y = 13$ (b) $x - y = 13$ (c) $2x + 3y = 13$ (d) $x^2 + y = 15$

Ans: (a) $x + y = 13$

80. The difference between the upper and lower limit is called

- (a) Group (b) Class size (c) Class interval (d) Class mark

Ans: (b) Class size

81. What is the median of the data 46, 64, 87, 41, 58, 77, 35, 90, 55, 33, 92?

- (a) 87 (b) 77 (c) 58 (d) 60.2

Ans: (c) 58

82. In the class - interval 70-80, 80 is the

- (a) Lower limit (b) Upper limit (c) Frequency (d) Range

Ans: (b) Upper limit

83. What is the mean of x, x + 3, x + 6, x + 9 and x + 12?

- (a) x + 3 (b) x + 6 (c) x + 9 (d) x + 12

Ans: (b) x+6

84. Which of the following is correct about mode?

- (a) It occurs most frequently (b) It is the average of the two middle terms
(c) It is central (d) It lies between the maximum and minimum observations

Ans: (a) It occurs most frequently.

85. Median of the data 13, 16, 12, 14, 19, 12, 14, 13, 14 is ____.

- (a) 14 (b) 19 (c) 12 (d) 13

Ans: (a) 14

86. Mode and median of the data 13, 16, 12, 15, 19, 12, 15, 13, 15 are:
(a) 13 & 15 (b) 15 & 13 (c) 15 & 15 (d) 19 & 13

Ans: (c) 15 & 15

87. _____ is based on all observations.

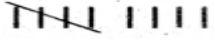
- (a) Range (b) Mean (c) Median (d) Mode

Ans: (b) Mean

88. The difference between maximum and minimum value of a data is called _____.

- (a) Frequency (b) Median (c) Mean (d) Range

Ans: (d) Range

89.  Denotes _____.

- (a) 5 (b) 10 (c) 9 (d) 8

Ans: (c) 9

90. Which of the following is true?

- (a) Range = Min. value – Max. value (b) Mode = Sum of the all observations 2
(c) Median is the middle most value of the data (d) Mean = Median

Ans: (c) Median is the middle most value of the data.

91. Mean of 11,12,13,14 is _____.

- (a) 9 (b) 10 (c) 11 (d) 13

Ans: (d) 13

92. The information collected is called _____.

- (a) Data (b) Range (c) Mean (d) None

Ans: (a) Data

93. Middle most value of a data is called _____.

- (a) Median (b) A.M. (c) Range (d) Total

Ans: (a) Median

94. The sum of 20 observations is 400, then the mean is _____.

- (a) 20 (b) 45 (c) 29 (d) 340

Ans: (a) 20

95. Median of numbers from 4 to 10 is _____.

- (a) 4 (b) 6 (c) 7 (d) 5.5

Ans: (c) 7

96. In a data Range = 100. Max. Value is 180, then minimum value = _____.

- (a) 110 (b) 180 (c) 90 (d) 80

Ans: (d) 80

97. Range of first 99 natural number is _____.

- (a) 100 (b) 98 (c) 90 (d) 1

Ans: (b) 98

98. Sum of 11 observation is 132, then the mean= _____.
(a) 11 (b) 16 (c) 12 (d) 19

Ans: (c) 12

99. Mode of 1, 1, 3, 7, 1, 2, 3, 1, 1, 4, 5, 6, 1 is _____.
(a) 4 (b) 7 (c) 3 (d) 1

Ans: (d) 1

100. Bimodal data has _____ modes.
(a) 2 (b) 4 (c) 3 (d) 1

Ans: (a) 2